

Course outline (Syllabus of the course)

Course Code: CS1109 Name of the Programme: BCA (Hons) Name of the Course: Computer Organization Semester: II		
Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
3	45 + 0 + 0	45
Course Objectives:		
a) Understand the fundamental components and operations of computer systems.		
b) Analyze instruction sets and addressing modes.		
c) Explore memory hierarchy, storage systems, and input/output mechanisms, including buses and I/O interfacing.		
d) Comprehend basic concepts of instruction pipelining and overall system performance enhancement techniques.		

Module - 1	9 Hrs
FUNDAMENTAL BLOCKS OF COMPUTER: Computer Types, Functional Units, CPU, I/O subsystems, memory, control unit COMPUTER ARITHMETIC: Signed number representation, Fixed and Floating point representations.	
Module - 2	9 Hrs
INSTRUCTION SET ARCHITECTURE: Memory Location and addresses, Memory Operations, Instruction format, addressing modes, assembly languages, Complex Instruction Set Computer (CISC), Reduced Instruction Set Computer (RISC), CISC vs RISC.	
Module - 3	9 Hrs
MEMORY HIERARCHY: Concept of hierarchical memory organization, Cache Memory- Mapping Functions and Replacement Algorithms, Examples of Mapping Techniques, Cache Coherence. INPUT OUTPUT AND SECONDARY STORAGE: I/O interface, Programmed IO, Memory Mapped IO, Interrupt Driven IO, DMA, Secondary Storage.	

Module - 4	9 Hrs
ALU, DATA-PATH AND CONTROL UNIT: Control Unit and design, Hardwired v/s Micro-programmed Control Unit, Horizontal micro-programmed Vs Vertical micro-programmed control unit, Synchronous Data Transfer, Asynchronous Data Transfer	
Module - 5	9 Hrs
INSTRUCTION PIPELINING: Pipelining, Types of pipelining, Pipelining - Execution, Stages and Throughput, Pipelining - Dependencies and Data Hazard, Branch Delays, Pipelining - Types and Stalling Different Instruction Cycles, Performance of Computer.	

Course Outcomes:
a) Describe the functional organization of a computer system and perform arithmetic operations using signed, fixed-point, and floating-point number representations.
b) Analyze instruction set architectures, addressing modes, and instruction formats, and distinguish between CISC and RISC architectures.
c) Evaluate memory hierarchy concepts, cache mapping techniques, replacement algorithms, cache coherence, and various I/O and secondary storage mechanisms.
d) Explain the design and functioning of ALU, datapath, and control units, and analyze pipelined execution, hazards, and performance metrics of computer systems.

Textbooks	
1	Carl Hamacher, Zvonks Vranesic, SafeaZaky (2002), Computer Organization, 6th edition, McGraw Hill, New Delhi, India. ISBN: 9789355323729

Reference Books	
1	M. Mano, Computer System Architecture, 3rd Ed., Pearson, 1992. ISBN: 9789332585607
2	William Stallings (2010), Computer Organization and Architecture- designing for performance, 8th edition, Prentice Hall, New Jersey. ISBN: 9788131732458

Reference Links

1	https://www.geeksforgeeks.org/computer-organization-and-architecture-tutorials/
2	https://www.javatpoint.com/computer-organization-and-architecture-tutorial
3	https://www.studytonight.com/computer-architecture/